CLASS 10 — MATHS · FORMULA SHEET

Chapter 1: **Real Numbers** | Quick revision, one page

Fundamental Theorem of Arithmetic

- Every composite number can be expressed as a product of primes, and this factorisation is unique apart from the order of the primes.
- Example: $360 = 2^3 \times 3^2 \times 5$.

HCF and LCM by prime factorisation

- HCF = product of the smallest power of each common prime factor.
- LCM = product of the greatest power of each prime factor present in the numbers.
- For two positive integers: $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$ (does not hold for three numbers).
- Co-prime numbers: HCF = 1 and LCM = product.

Irrational numbers

- $\sqrt{2},\ \sqrt{3},\ \sqrt{5}$ and, in general, $\sqrt{p}$ for a prime p are irrational.
- Rational $\pm$ irrational = irrational; non-zero rational $\times$ irrational = irrational.
- Sum, difference, product or quotient of two irrationals may be rational or irrational (e.g. $\sqrt{2} \times \sqrt{2} = 2$).

Decimal expansions of rational numbers

- $\frac{p}{q}$ in lowest terms terminates if $q = 2^m 5^n$ $(m, n \geq 0)$.
- Otherwise the decimal expansion is non-terminating and repeating.
- Irrational numbers have non-terminating, non-repeating expansions.

Method to prove $\sqrt{p}$ irrational

- Assume $\sqrt{p} = \frac{a}{b}$ with a, b co-prime; then $a^2 = pb^2$, so p divides a; write $a = pc$ and show p divides b; contradiction with co-primality.

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